V-T-301D
April 22, 1971
SUPERSEDING
Fed. Spec. V-T-301C
March 17, 1965

FEDERAL SPECIFICATION

THREAD, SILK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for silk thread used for machine and hand sewing.

1.2 Classification.

1.2.1 Types and classes. The thread shall be of the following types and classes as specified (see 6.1):

Type I - Machine twist (soft finish).

Type II - Buttonhole twist (hand).

Class 1 - Tailors (soft or waxed finish).

Subclass A - Soft.
Subclass B - Waxed.

Class 2 - Dry goods (soft finish).

Type III - Hand sewing, pound goods (waxed finish).

Type IV - Embroidery (soft finish).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-50 - Packaging and Packing of Thread for Domestic and Overseas Shipment.

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Federal Standard:

Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, Mo, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

The Color Association of the United States, Inc.

Department of Defense Standard Shades for Sewing Threads 1968

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York, N.Y. 10016.)

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3. REQUIREMENTS

3.1 The requirements specified in 3.8, 3.9, 3.10, section 5 and related provisions of section 4 apply only to thread purchased directly by the Government. All other requirements apply both to thread purchased by a supplier as a component for an end item and to thread purchased directly by the Government.

3.2 Standard sample. When a standard sample is established (see 6.2), the thread shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard is referenced.

3.3 Material.

3.3.1 Silk. The thread shall be made from domesticated silk (see 6.4) which shall be thoroughly degummed, free from loading or weighting, and twisted into thread to meet the requirements of this specification.

3.4 Color. The color (shade) shall be as specified (see 6.1) and shall conform to an approved shade standard (see 6.2), when tested as specified in 4.2.4. Shade designations by letter (e.g. - Olive Drab S-1) and related cable numbers (e.g. - CA 66022) refer to the DOD Standard Shades for Sewing Threads 1968 (see 2.2).

3.4.1 Colorfastness. The colorfastness of dyed finished thread shall be as specified in the applicable end item specification or as set forth in the contract or order (see 6.1), except that when requirements are not stated or referenced to a standard sample for colorfastness, the thread shall show "good" fastness to laundering, wet drycleaning and light as specified. When a standard sample for colorfastness is referenced the thread shall be equal to or better than the standard sample for the specified colorfastness properties or (when specific properties are not specified) for colorfastness to laundering, wet drycleaning and light. Tests for colorfastness (including tests for perspiration) when specified, shall be as specified in 4.2.4 as applicable.

3.5 Finish. The thread shall have a soft or wax finish as indicated for the type and class as specified in the applicable end item specification or procurement document (see 4.2.4). The finish shall conform to the requirements of 3.5.1 and 3.5.2.

3.5.1 Soft finish. Soft finish thread shall contain only the minimum amount of lubricant to facilitate sewing.

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3.5.2 Wax finish. Wax finish thread shall be a dressed thread which possesses a smooth surface and body suitable for hand sewing.

3.5.3 Finishing materials. No chemical finishes or treatments other than those specified shall be used without prior approval of the contracting officer.

3.5.4 Sewing finishes. Only beeswax paraffin, petroleum waxes and dimethyl-polysiloxanes or combinations thereof commonly used shall be permitted as sewing finishes unless prior approval is obtained from the contracting officer.

3.6 Direction of twist. Unless otherwise specified (see 6.1), the direction of final twist shall be "Z" twist, except for type III and type IV thread which shall be "S" twist when tested as specified in 4.2.4.

3.6.1 Elongation. The elongation of the thread shall be not less than 10 percent when tested as specified in 4.2.4.

3.7 Physical requirement. The finished thread shall conform to the applicable requirements specified in table I through IV for the number of plies, length per ounce, and breaking strength when tested as specified in 4.2.4.

TABLE I. Type I - Machine twist

Ticket number	Ply	Minimum length per ounce	Minimum breaking strength
		Yards	Pounds
00	3	1300	2.0
0	3	1150	2.4
A	3	900	3.0
B	3	730	3.7
C	3	550	5.0
D	3	460	5.9
E	3	330	8.2
F	3	235	10.9
FF	3	210	12.8

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TABLE II. Type II - Buttonhole twist

Ticket number	Ply	Minimum length per ounce after wax removed	Minimum breaking strength
		Yards	Pounds
Class 1 - Tailors			
6	3	170	15.2
8	3	210	12.8
10	3	235	10.9
12	3	330	8.2
Class 2 - Dry Goods			
D	3	460	5.9

TABLE III. Type III - Hand sewing

Ticket number	Ply	Minimum length per ounce after wax removed	Minimum breaking strength
		Yards	Pounds
A	2	850	3.0
C	2	520	5.0

TABLE IV. Type IV - Embroidery

Ticket number	Ply	Minimum length per ounce	Minimum breaking strength
		Yards	Pounds
BB	2	650	4.0
D	2	460	5.9
E	2	330	8.2

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3.7.1 Maximum length per ounce. The maximum length per ounce of any stated ticket number of thread shall be the specified minimum of the next finer size in the same type and number of plies. The maximum length per ounce of the finest sizes of threads listed in each group shall be not more than 25 percent above the minimum specified.

3.8 Knots in thread. Sizes FF and finer thread shall average not more than one thread knot per ounce of thread per holder; sizes heavier than FF shall average not more than one thread knot per two ounces of thread per holder.

3.9 Put-up. Unless otherwise specified (see 6.1), the thread shall be put-up on yardage or weight per holder basis on commercial spools, cones, tubes, or skein (hereinafter referred to as holders) as specified in the contract or order in accordance with applicable schedules of tables V through VIII. The average length or weight (as applicable) per holder shall be not less than the specified nominal amount minus 3.0 percent tolerance when examined (on a 10-holder basis) as specified in 4.2.2. The thread shall be put-up in one continuous length per holder and shall be so wound that each turn and layer is free from entanglement.

TABLE V. Put-up for type I - Machine twist

Ticket number	Yards per single headed spool	Yards per single headed tube	Yards per single headed spool	Yards per cone
00	925	3,700	5,400	10,600
0	800	3,200	4,800	9,400
A	630	2,520	3,800	7,400
B	520	2,080	3,120	6,040
C	410	1,640	2,400	4,600
D	345	1,380	2,040	3,880
E	245	980	1,520	2,840
F	175	690	1,140	2,100
FF	155	620	1,040	1,880

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TABLE VI. Put-up for type II - Buttonhole twist

Ticket number	Class 1, soft finish yards per spool 1/	Class 1, waxed finish yards per skein	Class 2, soft finish yards per spool 2/
6	24	170	-
8	28	210	-
10	33	235	-
12	44	330	-
D	-	-	10

1/ Type II thread, ticket number 6-12 to be braided 8 strands before spooling.

2/ Ticket number "D" is not to be braided.

TABLE VII. Put-up for type III - Hand sewing

Ticket number	Waxed finish, yards per skein
A	850
C	520

TABLE VIII. Put-up for type IV on cones

Ticket number	Net weight (minimum) ounces
BB-D-E	4 or 8

3.9.1 Braiding. Type II thread, except letter size No. D, shall be braided eight strands before spooling.

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3.10 Identification marking. Except when commercial identification markings are specified, each holder shall have a label attached in such a manner as to remain in place and be clearly legible until all thread has been removed. The label shall be printed with the information as specified below:

Stock number
Net weight (oz. or lbs.) or length (yards)
Direction of twist
Color
Ticket number and ply
Type and class
Nomenclature

3.11 Workmanship. The finished thread shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 End item examination.

4.2.1.1 Thread on the holder. The thread shall be examined on the holder and the visual defects listed in table IX shall be counted. The sample unit shall be one holder and the lot size shall be expressed in units of one holder each. The AQL shall be 4.0 defects per 100 units and the inspection level shall be S-3 of MIL-STD-105.

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TABLE IX. Defects of thread on holders

Examine	Defect
Identification label	Missing, incorrect, incomplete, illegible, or insecurely attached.
Type of holder	Other than specified.
Surface condition	Loose ply, cut, tear, chafe, slip, affecting strength of thread or interfering with easy location of end and initial unwinding.
Color	Other than specified. Uneven-apparent on successive layers or end to end.
Cleanness	Dirt, spot or stain clearly noticeable. 1/
Finish	Other than specified. Uneven, lumpy, bare or thin spot.
Braiding	Not braided when specified.

1/ At normal inspection distance approximately 3 feet.

4.2.1.2 Thread as unwound. The thread shall be examined by unwinding, and defects listed in table X shall be counted regardless of their proximity to each other. The sample unit shall be one holder and the lot size shall be expressed in units of one holder each. The AQL shall be 4.0 defects per 100 units and the inspection level shall be S-3 of MIL-STD-105.

TABLE X. Defects of thread as unwound

Examine	Defect
Continuous length	Not in continuous lengths.
Knots	Size FF and finer thread averages more than one thread knot per ounce of thread. Size FF and heavier thread averages more than one thread knot per two ounces of thread.

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TABLE X. Defects of thread as unwound (cont'd)

Examine	Defect
Winding	Improperly or not firmly wound, resulting in kinks, knots, entangling or slippage during unwinding or otherwise affecting free underwinding of thread.
Tackiness or adhesion	Strands adhere to each other or to holder affecting unwinding tension.

4.2.2 Examination for length and net weight. The sample unit shall be 10 wound holders, taken at random from the lot under test.

4.2.2.1 Length per holder. Ten holders of each lot shall be measured individually. Measurements shall be made over an accurate measuring clock under normal winding tensions. The values obtained shall be averaged and this average shall represent the length per holder in the lot. The lot shall be unacceptable if the length per holder is less than the amount specified minus the 3.0 percent tolerance (see 3.9).

4.2.2.2 Net weight per holder. Ten holders of each lot shall be weighed individually. The weight of thread obtained shall be averaged and this average shall represent the net weight of thread per holder in the lot. The lot shall be unacceptable if the net weight per holder is less than the amount specified minus the 3.0 percent tolerance (see 3.9).

4.2.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-50 to determine that packaging, packing and marking complies with the section 5 requirements.

4.2.4 Testing of the end item. The methods of testing specified in Fed. Std. No. 191 wherever applicable and as listed in table XI shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes. All test reports shall contain the individual values utilized in expressing the final results. The AQL for each characteristic shall be 6.5 test failures per 100 units and the inspection level shall be S-1. The sample unit shall be one wound holder. The lot size shall be expressed in unit of one holder.

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TABLE XI. Test methods

Characteristic	Requirement paragraph	Test method
Silk	3.3.1	1110 1/
Shade	3.4	4.2.4.1
Colorfastness to:		
Laundering	3.4.1	5614
Light	3.4.1	5660
Wet drycleaning	3.4.1	5622
Perspiration	3.4.1	5680
Finishes	3.5	Visual 1/
Direction of final twist	3.6	4050
Elongation	3.6.1	4100 2/
Number of plies	3.7	Visual 3/
Breaking strength	3.7	4100 2/
Length per ounce	3.7	4010 4/ 5/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Five specimens shall be tested for each sample unit.

3/ One determination and the result reported as "pass" or "fail".

4/ One skein shall be measured for each sample unit.

5/ The wax shall be removed from types II and III thread in accordance with the chloroform soluble material procedure of method 2611 and the extracted thread shall be oven dried at a temperature of 79° to 81°C, until free of solvent.

4.2.4.1 Shade and colorfastness evaluation.

4.2.4.1.1 Procedure for removal of the sewing finish from all thread and waxes from types II and III thread when specified. Before evaluation for shade matching, and testing for colorfastness, except colorfastness to wet drycleaning, the sample of thread shall be wet drycleaned in accordance with method 5622 of Fed. Std. No. 191, (disregarding reference to the standard sample). Excess solvent shall be removed by centrifuging or wringing. The sample shall then be rinsed in distilled water at 120° to 160°F, and dried at a temperature not exceeding 180°F. The dried sample shall then be conditioned for a minimum of 4 hours prior to evaluation for shade match. Colorfastness tests shall be as specified in table XI.

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4.2.4.1.2 Matching. The sample shall be wound or prepared for examination in a manner similar to that for the standard shade sample. The sample and the standard shall be compared under natural (north sky) daylight or under an artificial daylight lamp having a color temperature of 7500° Kelvin. The sample shall be unacceptable if it is not a good match to the shade standard. The sample shall also be compared to the standard under incandescent lamplight at 2800° Kelvin. The sample shall be unacceptable if it is not a good approximation to the shade standard.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.2).

5.1.1 Levels A, B and C. The thread shall be packaged in accordance with the applicable requirements of PPP-P-50.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Levels A, B and C. The thread shall be packed in accordance with the applicable requriements of PPP-P-50.

5.3 Marking. In addition to any special marking specified in the contract or order, interior packages and shipping containers shall be marked in accordance with applicable provisions of PPP-P-50.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, class, and nomenclature of thread required (see 1.2).
(c) Subclass required (applicable to type II thread only) (see 1.2).
(d) Color (shade) required (see 3.4).
(e) Colorfastness properties required (see 3.4.1).
(f) Direction of twist, if other than specified (see 3.6).
(g) Put-up required (see 3.9).
(h) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 Standard sample. For military shades, the standards for shade reference are those contained in the Standard Color Card of Official Standardized Shades for Sewing Threads (see 2.2). For other shades, address the procuring office issuing the invitations for bids or request for proposal. Standard samples frequently used by the military (directly or for components of end items) have been made available to normal suppliers as basic reference for the establishment of the supplier's secondary standards conforming to the characteristics of the official standards. The basic shade standards are those appearing in the DOD Color Card. Prime thread manufacturers who inadvertently have not been furnished standards for establishment of their colorfastness standards in accordance with this specification may obtain samples for this purpose from the U. S. Army Natick Laboratories, Natick, Massachusetts 01760, ATTN: AMXRE-CTR.

6.3 Colorfastness. Normally colorfastness requirements for thread are to be in keeping with those specified for the basic material into which the thread will be sewed (see 3.4.1).

Typical examples are:

Use	Colorfastness requirement
Thread for sewing wool clothing which is dry cleaned.	Good colorfastness to wet dry-cleaning, light and perspiration.
Thread for sewing wool clothing which is laundered.	Good colorfastness to laundering, light and perspiration.

6.4 Silk (see 3.3.1). Domesticated silk is defined for the purposes of this specification as that silk originating from domesticated or artificially cultivated silk worms.

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MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
Air Force - 45, 82

User activities:

Army - MU
Navy - MC, OS

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

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